

AYUSH GHOSH

U.S. Citizen

☎ 443-917-8321 ✉ ayushrghosh@berkeley.edu [in](#) [LinkedIn](#) [globe](#) [Portfolio](#)

Education

University of California, Berkeley

Expected May 2028

Mechanical Engineering, B.S.; Minor in Electrical Engineering and Computer Science

GPA: 3.95

- **Relevant Coursework:** Engineering Design and Development, Computer-Integrated Manufacturing, Three-Dimensional Modeling and Design, Mechanics, Thermodynamics, Introduction to Circuits and Devices, Electricity and Magnetism

Experience

FSAE Formula Electric, UC Berkeley

January 2026 – Present

Accumulator Mechanical Engineer

Berkeley, CA

- Co-designed the team's custom **140S3P** lithium-ion high-voltage battery pack from concept through fabrication, driving GD&T tolerance stack-ups, materials selection for dielectric insulation, thermal management, and design for manufacturability and assembly (DFMA)
- Instrumented a validation test fixture with an Arduino-based thermocouple array, four-point resistance measurement, and **Python** data logging to verify **ANSYS** thermal simulations, root-causing hotspots that drove busbar redesigns and cut accumulator mass by **18%**
- Developed a waterproof, electrically insulated HV/LV enclosure with a silicone gasket seal for environmental protection, and packaged cell-monitoring PCBs into the module housing alongside the electrical team

KwikKart

May 2026 – August 2026

Hardware Engineering Intern

Austin, TX

- Re-engineered a smart tablet's rear structural assembly in **SolidWorks**, packaging electronics, cameras, and a spring-loaded, tool-free battery ejection mechanism while cutting assembly mass by **30%**
- Prototyped and validated forced-air thermal management for an NVIDIA Jetson AGX Orin, iterating a CNC-machined heatsink and chimney flow path in **ANSYS CFD** to meet tight packaging constraints
- Authored **KiCad** schematics of the tablet's full electronic architecture, streamlining mechanical–electrical integration and debugging

Volchor

September 2025 – May 2026

Mechanical Engineering Intern

San Francisco, CA

- Analyzed and qualified structural components in **ANSYS** under combined aerodynamic and thermal loading, applying factors of safety and material allowables for a high-acceleration flight environment
- Wrote **MATLAB** physics models of payload dynamics (stability, transient loads, kinematics) to inform structural sizing and detachable heat-flux-absorbing shell concepts
- Modeled vehicle geometry and supporting vacuum infrastructure in **SolidWorks**, including pressure boundaries, isolation gates, and pumps

Projects

Custom Quadcopter

June 2026 – August 2026

Personal Project

- Sized motors, propellers, ESCs, and battery from first principles and validated a **~0.9 kg SolidWorks** airframe with FEA, achieving a **~3.6:1** thrust-to-weight ratio and **~12 min** flight time
- Laid out a **6-layer** STM32F405 flight-controller PCB in **KiCad** (IMU, barometer, RC receiver) and wrote firmware from scratch for sensor fusion, attitude stabilization, and motor control
- Fabricated a tethered flight-test fixture and tuned PID gains to a stable hover within $\pm 2^\circ$ roll/pitch under disturbances

Computed Axial Lithography 3D Printer

July 2026 – Present

Design for Nanomanufacturing Lab, UC Berkeley

- Building HeliCAL60, a benchtop computed axial lithography (CAL) 3D printer that cures full parts volumetrically by projecting tomographic light patterns into a rotating vial of photopolymer resin
- Designing the printer's positioning mechanisms in **SolidWorks**, including mounts and structural interfaces for **5 μ m-resolution** motorized translation stages and a lead-screw drive for repeatable axial positioning
- Implementing a closed-loop alignment system that pairs a CMOS image sensor with the motorized stage, using real-time feedback to auto-correct optical/rotational misalignment

Skills

Design & Simulation: SolidWorks, CATIA V5, Siemens NX, PTC Creo, Fusion 360, Onshape, ANSYS, FEA, CFD, Modal Analysis

Fabrication & Test: GD&T, DFMA, CNC Machining, CAM, Sheet Metal, 3D Printing, Test Fixture Design, Data Acquisition

Hardware + EE Tools: KiCad, Altium Designer, LTspice, Microcontrollers (STM32, Arduino, Raspberry Pi)

Software & Languages: C/C++, Python, MATLAB, Java, Bare-Metal Firmware